

MOHD ANAS

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EDUCATION

B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Aug 2027

Coursework: OS concepts, Embedded Systems, Data Structures & Algorithms, Object Oriented Programming, Backend Systems, Machine Learning, Deep Learning

PROFESSIONAL EXPERIENCE

Aarogya Kaya | SDE Intern | Remote

July 2025 - Jan 2026

- **Architected Go Backend services** for Shiprocket and Instagram Graph API integrations, implementing CQRS architecture and a three-tier caching strategy (in-memory, Redis, and database), reducing average read latency from **420ms to 38ms (91%)** and maintaining **p95 latency below 250ms**.
- Refactored 15+ Backend modules using **SOLID principles** and **Design Patterns** (such as **Dependency Injection**), reducing code duplication by 30% and improving feature delivery speed.
- Engineered fault-tolerant asynchronous synchronization pipelines with retry mechanisms and **idempotent processing**, supporting **500+ daily events** while **99.9%** data consistency.

Open Source Contributor | Remote

July 2025 - Present

- **Contributed to Next.js (Vercel)** by migrating the patch-next utility from CommonJS to TypeScript ESM, eliminating repeated dependency reinstalls and reducing contributor setup overhead.
- Contributed **22 merged Pull Requests** and opened **12 Issues** across open-source projects, fixing bugs, implementing features, and improving developer tooling.

PROJECTS

a3.store | Distributed Key-value Store | Python, EKS, AWS, Kubernetes, Prometheus, Security

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- Developed a leaderless **Distributed key-value store** implementing replication, failure detection, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed on AWS EKS using Kubernetes StatefulSets and persistent volumes, enabling **rolling upgrades, pod recovery** and **self-healing behavior**.
- Secured inter-node communication using **mTLS** and implemented role-based access controls for administrative operations.

Scull Driver | Linux Character Device Driver | Rust, Linux Kernel

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- Reimplemented the Linux `scull` character device driver in Rust using **Rust-for-Linux** APIs, implementing device lifecycle, read/write interfaces, and `ioctl` operations.
- Implemented **kernel-space data management** using `KBox/KVec`, dynamic `qset/quantum` storage, and **mutex-based synchronization** for safe concurrent device access.
- Built and loaded the out-of-tree `.ko` module against a **Rust-enabled Linux kernel** using `Kbuild`, gaining hands-on experience with **Linux kernel** and **device-driver development**.

Shop-Intelligence | Real-Time Customer Behavior Analytics | Python, Computer Vision, Queue, Pytest

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- Built a distributed real-time **computer vision pipeline** for four synchronized CCTV streams using concurrent workers and message queues, **processing 15 FPS** while reducing frame-processing latency to **50 ms**.
- Designed an **event-driven** analytics engine and global visitor registry that generated real-time backend events for customer behavior dashboards and retail insights.

TECHNICAL SKILLS

Languages: Python, Go, C/C++, TypeScript, Javascript, SQL

Backend & System: FastAPI, Node.js, gRPC, REST APIs, Kafka, Redis, Distributed Systems

Cloud & Infrastructure: AWS, EKS, Kubernetes, Docker, Terraform, Linux, EC2, VPC, Terraform

Security & monitoring OAuth 2.0, JWT, IAM, mTLS, RBAC, Prometheus, Grafana, Loki

ACHIEVEMENT

- Ranked **3,026th** among **17,000+** participants in Codeforces Round 1110 (Div. 1 + Div. 2)
- **Solved 500+ DSA problems** across Codeforces, LeetCode, CodeChef and GeeksforGeeks.